$x_1 + 2(-2x_3 + 3x_5) - 5x_3 - x_5 = 0$

$\Rightarrow x_1 = 9x_3 - 5x_5.$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 9 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} -5 \\ 3 \\ 0 \\ -2 \\ 1 \end{bmatrix} x_5.$$

The bases for Nul A: $\left\{ \begin{bmatrix} 9 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$

Dimension: 2

In Exercises 13 and 14, find a basis for the subspace spanned by the given vectors. What is the dimension of the subspace?

13. $\begin{bmatrix} 1 \\ -3 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} -3 \\ 9 \\ -6 \\ 12 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} -4 \\ 5 \\ -3 \\ 7 \end{bmatrix}$

Solution: The basis of a subspace represents a set of linearly independent vectors which spans the subspace.

For the set of vectors to be linearly independent, the matrix equation $Ax = 0$ must have only the trivial solution.

$$A = \begin{bmatrix} 1 & -3 & 2 & -4 \\ -3 & 9 & -1 & 5 \\ 2 & -6 & 4 & -3 \\ -4 & 12 & 2 & 7 \end{bmatrix}.$$

By Theorem 3 from Chapter 1, the matrix eqn. $Ax = 0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A\ 0]$.

$$\begin{bmatrix} 1 & -3 & 2 & -4 & 0 \\ -3 & 9 & -1 & 5 & 0 \\ 2 & -6 & 4 & -3 & 0 \\ -4 & 12 & 2 & 7 & 0 \end{bmatrix} \xrightarrow[+3\times \text{Row}1]{\text{Row}2 = \text{Row}2} \begin{bmatrix} 1 & -3 & 2 & -4 & 0 \\ 0 & 0 & 5 & -7 & 0 \\ 2 & -6 & 4 & -3 & 0 \\ -4 & 12 & 2 & 7 & 0 \end{bmatrix} \xrightarrow[2\times \text{Row}1]{\text{Row}3 = \text{Row}3 -}$$

$$\begin{bmatrix} 1 & -3 & 2 & -4 & 0 \\ 0 & 0 & 5 & -7 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ -4 & 12 & 2 & 7 & 0 \end{bmatrix} \xrightarrow[+4\times \text{Row}1]{\text{Row}4 = \text{Row}4} \begin{bmatrix} 1 & -3 & 2 & -4 & 0 \\ 0 & 0 & 5 & -7 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 10 & -9 & 0 \end{bmatrix} \xrightarrow[-2\times \text{Row}2]{\text{Row}4 = \text{Row}4}$$